

What is Data Engineering?

Data Engineering refers to the practice of designing and building systems for the aggregation, storage and analysis of data at scale. Data engineers primarily work with algorithms, data pipelines that sort raw data into industry ready datasets.

Use Cases

- Data Collection, Storage and Management
- Real-Time Data Analysis
- Machine Learning
- Deep Learning for Complex Patterns

Difference between Data Engineers & Data Scientists

Data engineers and scientists go hand-in-hand but their primary workloads are different, with engineers focusing on building, and maintaining infrastructure for data-related processes, while data scientists focus on drawing information from data at hand, and generate insights, build predictive models, etc.

Top Data Engineer Skills Required

- **Data Collection and Integration** - Engineers collect data from sources such as databases, APIs, SAPs. They have to implement pipelines to ensure smooth operations on the data which flow from these various sources into the storage systems.
- **ETL (Extract, Transform, Load) Processes** - The building block of data engineering. Engineers have to design ETL pipelines to transform raw data into a format suitable for analysis.
- **Big Data** - Engineers should be familiar with big data technologies like Hadoop, Spark, etc. for massive datasets.
- **NoSQL Databases** - Familiarity with relational databases is not enough, engineers should have an idea about NoSQL databases like MongoDB and Cassandra, for unstructured and semi-structured data.
- **Cloud Computing** - Enterprise data engineering platforms require AWS, Azure, Google Cloud knowledge, especially for scalable and cost effective data solutions.
- **Distributed Systems** - Engineers should be able to design distributed architectures for distributed processing of large scale datasets, etc.

- **Real-Time Processing** - Data is often streamed from various sources, and engineers should be able to design systems to handle this influx of data.